

AMIR GHODRATI

Ph.D. Researcher · AI in Construction & VDC

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EDUCATION

Ph.D. in Civil Engineering

Jan 2025 – Present

University of Illinois Urbana-Champaign · Advisor: Prof. Mani Golparvar-Fard

- Research focus: AI-driven automation of construction progress monitoring using computer vision and deep learning
- Developing automated pipelines for semantic segmentation of construction element types (UNIFORMAT II) and installation stages from site photographs
- Integrating Building Information Modeling (BIM), photorealistic simulation, and synthetic dataset generation for training vision models
- Extending theoretical framework of 4D as-planned/as-built deviation detection to fine-grained, element-level progress recognition

Master of Science in Civil Engineering

Jan 2023 – Dec 2024

University of Illinois Urbana-Champaign · Construction Engineering and Management

Bachelor of Science in Civil Engineering

Aug 2018 – Aug 2022

Sharif University of Technology, Tehran, Iran · Ranked #1 engineering university in Iran

- Selected as Exceptional Talented student; ranked in the top 0.5% nationally in Iran's KONKUR university entrance examination

RESEARCH EXPERIENCE

Graduate Research Assistant

Jan 2023 – Present

Real-time and Automated Monitoring and Control (RAAMAC) Lab · UIUC Dept. of Civil Engineering and Computer Science

Advisor: Prof. Mani Golparvar-Fard

- Conducting research on automation of construction progress monitoring through integration of AI, computer vision, and Building Information Modeling (BIM)
- Developing automated synthetic dataset generation pipelines for training deep learning segmentation models on construction site imagery
- Applying semantic segmentation and object detection techniques to classify construction element types and installation stages from site photographs
- Building BIM-based workflows for virtual design and construction (VDC) progress tracking, linking as-planned 4D models to as-built visual data
- Leveraging photorealistic rendering environments and procedural 3D scene generation to produce labeled training data at scale

PROFESSIONAL EXPERIENCE

BIM/VDC/AI Innovation Intern

Jun 2025 – Aug 2025

EnTech Engineering · New York Metropolitan Area, Manhattan, NY · On-site

- Developed a 3D semantic segmentation pipeline using MMDetection3D to automate the Scan-to-BIM process for large-scale construction point cloud models
- Addressed challenges of high-density point cloud data by designing a modular segmentation approach, decomposing complex models into manageable, semantically labeled segments

- Delivered an AI-driven BIM workflow enabling VDC teams to extract structured building components from raw scan data with reduced manual effort

Graduate Teaching Assistant – CEE320

Aug 2023 – Present

Construction Engineering and Management · University of Illinois Urbana-Champaign

Appointment awarded competitively by the Department of Civil Engineering for three consecutive years

- Guide and mentor students through core construction engineering concepts including project planning, scheduling, and cost control
- Hold office hours, assist with coursework, and support instructors in delivering course content on construction methods and management practices

Virtual Design and Construction Engineering Intern

May 2021 – Sep 2021

Seism Ground Structure Consulting Engineers (SGS Co.) · Tehran, Iran

- Conducted daily site visits to monitor construction progress on government school renewal projects and bridge safety inspections in North Provinces of Iran
- Assisted in VDC workflows and progress control reporting

GRADUATE COURSEWORK

Machine Learning · Deep Learning · Computer Vision · Computational Photography · Construction Engineering and Management · Virtual Design and Construction

TECHNICAL SKILLS

AI & Machine Learning: Python, PyTorch, MMDetection3D, deep learning model design and training, semantic segmentation, object detection, Jupyter Notebook

BIM & VDC: Autodesk Revit, Navisworks, AutoCAD, Bluebeam, Solibri, Primavera P6, BIM-based progress monitoring workflows

3D & Simulation: NVIDIA Isaac Sim, Unreal Engine, Unity, 3ds Max, Graphics, synthetic data generation

Programming: Python, C, C++, Git

HONORS & AWARDS

Departmental Teaching Assistantship Appointment

UIUC Dept. of Civil Engineering, 2023–2026

Competitively awarded full-time TA appointment by the department for three consecutive academic years

Exceptional Talented Student Designation

Sharif University of Technology, 2018

Ranked in the top 0.5% nationally among all participants in Iran's Konkur university entrance examination

REFERENCES

Available upon request

Prof. Mani Golparvar-Fard — CSO, Reconstruct Inc. & Professor, University of Illinois Urbana-Champaign
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